

Machine Learning Notes

A simple study guide for testing a PDF-based RAG chatbot

1. What is Machine Learning?

Machine Learning is a branch of artificial intelligence that allows computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task. A machine learning system usually learns from examples and improves its performance as it receives useful data.

2. Supervised Learning

Supervised learning is a type of machine learning in which a model learns from labeled data. Each training example contains an input and a known correct output. The model learns the relationship between the inputs and outputs and then uses that relationship to make predictions on new data.

Two common supervised learning tasks are classification and regression. Classification predicts a category, while regression predicts a numerical value.

3. Unsupervised Learning

Unsupervised learning works with data that does not have labeled outputs. The goal is to discover useful patterns, structures, or groups within the data. Clustering is a common example. In clustering, similar data points are grouped together based on their characteristics.

4. Classification

Classification is a supervised learning task in which the model predicts a discrete category or class. For example, an email system can classify messages as spam or not spam. A classification model learns from examples where the correct class is already known.

5. Regression

Regression is a supervised learning task used to predict numerical values. For example, a regression model could predict the price of a house using features such as its size, location, and number of rooms. The predicted result is a number rather than a category.

6. Training and Testing Data

A dataset is often divided into training data and testing data. Training data is used to teach the machine learning model. Testing data is used to evaluate how well the trained model performs on data it has not seen during training.

7. Overfitting

Overfitting happens when a machine learning model learns the training data too closely, including details or noise that do not generalize to new data. An overfitted model may perform very well on training data but poorly on unseen data.

8. Neural Networks

A neural network is a machine learning model inspired by the way biological neurons process information. It is made of connected units called neurons, usually organized into layers. Neural networks can learn complex patterns and are widely used in areas such as image recognition, natural language processing, and speech recognition.